

José Santos

SENIOR DATA ENGINEER

📧 josemrsantos | 🌐 josemrsantos | 🏠 josemrsantos.com | 📍 London, UK

Senior Data Engineer with 15+ years of experience designing, building and improving Data Platforms, Warehouses, Data Lakes and Data Lake-houses.

Strong hands-on background in Python, SQL and AWS, with substantial experience across PostgreSQL and Redshift and recent work with Apache Iceberg. Experienced in ETL/ELT, data modelling, ingestion and orchestration, including building scalable AWS S3 data-lake solutions and near-real-time data pipelines. Has progressed through senior, staff, lead and management roles while remaining close to engineering delivery. Now focused on a hands-on Senior Data Engineer / individual-contributor role within an established data team.

Work Experience

Midnite | Gaming and sports betting startup

Remote - London

LEAD DATA ENGINEER

Apr. 2024 — Jul. 2026

- Delivered a **CDC** solution using Estuary for sub-second PostgreSQL data synchronisation, significantly improving data freshness
- Diagnosed and resolved a long-standing Fivetran asset testing issue in dbt, unblocking the team after months of friction
- Strengthened engineering documentation and pipeline maintainability, reducing knowledge silos
- Contributed to technical assessment and hiring of a senior data engineer

Speedcast | Global satellite communications and network services

Remote - London

DATA ENGINEERING MANAGER

Oct. 2024 — Jul. 2026

- Designed platform architecture using **Confluent Kafka Cloud** and **Onehouse** to stream data into **Apache Iceberg** tables via AWS Glue Catalog, supporting schema evolution and multi-engine access
- Built and managed the data team, hiring and mentoring Data Scientists, Data Engineers and Cloud DevOps engineers while remaining involved in technical architecture and delivery
- Took ownership of the DevOps engineering function following a departure, maintaining infrastructure-as-code deployments and security practices
- Translated business requirements into pragmatic data products across distributed teams and technology vendors

MSK.ai / MyRecovery.ai / HOPCo | Musculoskeletal healthcare data and AI platform

Hybrid - Central London

LEAD DATA ENGINEER

Oct. 2023 — Apr. 2024

- Led the data team while maintaining and planning the rebuild of the core **Data Warehouse** used for KPI reporting
- Improved near-real-time dashboards and optimised ETL infrastructure to support evolving reporting and operational requirements

WorldRemit / Zepz | Digital remittance and cross-border payments

Central London / Full remote since 2020

DATA ENGINEER / STAFF DATA ENGINEER / DATA ENGINEERING MANAGER

Oct. 2016 — Apr. 2023

- Promoted to Staff Data Engineer within 15 months; later managed a team of seven senior engineers at peak
- Led the migration from a PostgreSQL data warehouse to a scalable **AWS S3 data lake** with Python ingestion, improving analytical capability and Data Science velocity
- Worked extensively with AWS data engineering services and introduced **Airflow**, Docker, AWS Batch and EC2 Spot instances
- Balanced delivery with engineering quality through code reviews, testing, security controls and documentation
- Migrated the Redshift data warehouse to **Snowflake** and ran a dbt proof of concept
- Founding contributor to the Data Engineering, Fraud/ML and BI/Analytics teams

DUCO | Cloud data reconciliation software

Central London

SENIOR SOFTWARE ENGINEER

May 2016 — Oct. 2016 (6 months)

- Worked on a data reconciliation product for financial institutions, delivering assigned engineering work within a short engagement

Import.io | Web data extraction platform

Central London

LEAD PYTHON DEVELOPER (DATA TEAM)

Jan. 2015 — Apr. 2016

- Designed and built a JSON-config-driven data delivery framework using Python and EC2, later containerised with Docker, for customers including Penguin Random House UK and P&O Cruises. Owned architecture, scalability, testing and documentation

Education & Certifications

FCT/UNL (NOVA FCT)

BSC SOFTWARE ENGINEERING (5-YEAR DEGREE)

2002

Amazon Web Services

AWS CLOUD PRACTITIONER

- Credly badge

Skills

Core

Data Engineering, Data Warehouse / Data Lake / Lakehouse design, ETL/ELT, dimensional modelling (Kimball), 3NF

Languages

Python, SQL

Primary Data Platforms

PostgreSQL, Amazon Redshift, Apache Iceberg, Athena, Dremio, TimescaleDB

Additional Platform Experience

Snowflake, Onehouse

Orchestration & Pipelines

Airflow/MWAA, Dagster, Kafka/Confluent, Fivetran, Estuary; some hands-on experience with dbt

AWS & Infrastructure

S3, EC2, ECS, Batch, EMR, Lambda, Glue, CloudFormation, CloudWatch, Docker, IaC; some hands-on experience with Terraform, Step Functions, SQS and SNS

Engineering

Data ingestion, CDC, pipeline reliability, schema evolution, testing, documentation, observability

Tooling & Observability

DataDog, TeamCity, Jenkins

Selected Technical Projects

AWS data lake

Led migration from PostgreSQL DWH to an AWS S3 data lake with Python ingestion at WorldRemit / Zepz

Data Lakehouse

Implemented a Data Lakehouse and a streaming architecture using Confluent Kafka, Onehouse, Iceberg and AWS Glue Catalog at Speedcast

PostgreSQL CDC

Delivered sub-second PostgreSQL synchronisation using Estuary at Midnite

AI engineering projects

Built multi-agent and LLM-assisted engineering tooling; GitHub portfolio available at github.com/josemrsantos. Some development also made in production environments at Speedcast and Midnite

Communication & Collaboration

- Fluent in English and Portuguese
- Experienced at translating data engineering and architecture concepts into clear business language for non-technical stakeholders
- Extensive experience working with near-shore, offshore and cross-timezone teams

Interests

Board games; previously organised a regular company lunchtime group that brought colleagues together across teams.